

DAILY UPDATE: MATH 101: LINEAR & MULTILINEAR ALGEBRA (FALL 2026)

JULIETTE BRUCE

CONTENTS

1. Week 1	1
1.1. Monday, September 14, 2026	1
1.2. Wednesday, September 16, 2026	2
2. Week 2	2
2.1. Monday, September 21, 2026	2
2.2. Wednesday, September 23, 2026	2
3. Week 3	3
3.1. Monday, September 28, 2026	3
3.2. Wednesday, September 30, 2026	3

1. WEEK 1

1.1. Monday, September 14, 2026. Our first class! This course is a modern introduction to the study of linear and multilinear algebra. A broad theme throughout this class will be that we understand an algebraic object not just through its elements, but through the maps into and out of it and the quotients we can form from it; linear algebra over a field then appears as one especially nice case of a more general story about rings and modules. We worked on Worksheet 1.1, which introduces rings, ring homomorphisms, and ideals. We began by deriving basic consequences of the ring axioms, such as the uniqueness of identities and inverses, the identity $a0 = 0$, and the fact that the units $R^\times$ form a group, and then looked at first examples including $\mathbb{Z}/n\mathbb{Z}$, polynomial rings $\mathbb{K}[x]$, and matrix rings $\text{mat}_n(\mathbb{K})$, paying attention to units, zero divisors, and noncommutativity (e.g., $E_{12}E_{21} \neq E_{21}E_{12}$). We then defined ring homomorphisms together with their kernels and images, and observed that kernels are two-sided ideals. This led to the construction of the quotient ring R/I , its universal property, and the First Isomorphism Theorem for rings. As applications we showed that $\ker(\text{ev}_a) = \langle x - a \rangle$, so that $\mathbb{K}[x]/\langle x - a \rangle \cong \mathbb{K}$, and studied the ring of dual numbers $\mathbb{K}[\varepsilon] = \mathbb{K}[x]/\langle x^2 \rangle$ as a first example of a ring with a nonzero nilpotent element. The worksheet concludes with ring automorphisms (including $\text{Aut}_{\mathbb{R}}(\mathbb{C})$ and $\text{Aut}_{\mathbb{K}}(\mathbb{K}[x])$), a proof that $\langle 2, x \rangle \subset \mathbb{Z}[x]$ is not principal, and further examples such as formal power series, endomorphism rings, group rings, and the Weyl algebra.

1.2. Wednesday, September 16, 2026. We worked on Worksheet 1.2, which introduces modules, the central objects of this course. A module over a ring R is an abelian group with a compatible action of R , and so generalizes a vector space by allowing the scalars to come from any ring. We saw that $\mathbb{K}$ -modules are exactly $\mathbb{K}$ -vector spaces, that every abelian group has a unique $\mathbb{Z}$ -module structure, and that R and its left ideals are R -modules. After defining submodules, R -linear maps, kernels, and images, we developed a key dictionary: a $\mathbb{K}[x]$ -module is the same thing as a vector space V together with a linear operator T , where submodules correspond to T -invariant subspaces and module homomorphisms to intertwining maps. We explored this through a nilpotent operator on $\mathbb{K}^2$, whose module V_T turns out to be isomorphic to the dual numbers $\mathbb{K}[\varepsilon]$ from Monday. We then constructed quotient modules M/N , proved their universal property, and established the First, Second, and Third Isomorphism Theorems along with the Correspondence Theorem, working through examples such as $4\mathbb{Z}$ and $6\mathbb{Z}$ inside $\mathbb{Z}$. The worksheet concludes by computing $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z}) \cong \mathbb{Z}/\gcd(m,n)\mathbb{Z}$ and by showing that an R -module structure on an abelian group M is the same thing as a ring homomorphism $R \rightarrow \text{End}_{\mathbb{Z}}(M)$.

2. WEEK 2

2.1. Monday, September 21, 2026. We took our first quiz and worked on Worksheet 2.1, in which we started developing the language of category theory, which supplies a rich universal approach to discussing many of the properties of rings, modules, and vector spaces all at one. From this perspective we work with objects and morphism instead of elements, and as such constructions are not specified by describing the elements of an object, but instead by describing the maps into or out of it. We began with the definition of a category and collected the examples we have already met, the worksheet has examples that we did not cover in class of other examples. We asked what injectivity and surjections look like when only morphisms are available, defining monomorphisms and epimorphisms by cancellation. We saw that for sets and vector spaces monomorphisms are the same as injections and epimorphisms the same as surjections. However, this is not true in general. Next came functors, both the forgetful ones and duality, which reverses arrows and so is a functor on the opposite category. With this vocabulary in place we turned to universal properties proper: free vector spaces, products and co-products. Using free vector spaces we defined bases for vector spaces, and on the worksheet you will check that Zorn's lemma implies every vector space has a basis. We then talked a bit about exact sequences of vector spaces, short exact sequences, the splitting lemma and rank nullity. **Reminder you have a take-home quiz due via Canvas on Friday.**

2.2. Wednesday, September 23, 2026. We worked on Worksheet 2.2. We went through co-products and products in the category of vector spaces with the first being given by direct sum and the later by cartesian product. Notice these agree when we are taking a finite product/co-product, but not in general. Next generalizing duality we showed that $\text{Hom}_{\mathbb{K}}(U, -)$ and $\text{Hom}_{\mathbb{K}}(-, U)$ are functors. More than that they are very special types of functors in that they each preserve short exact sequences of vector spaces. (This used the

splitting lemma in a crucial way.) We call such functors exact. The second half of the worksheet constructed the tensor product $V \otimes W$ as a free vector space on $V \times W$ modulo the bilinearity relations, established its universal property and uniqueness. We also discussed some of the basic properties of tensor products. Finally we proved the tensor–Hom correspondence. On the worksheet you will see how this is related to a categorical concept called adjunction, and will show that $W \otimes V^\vee \cong \text{Hom}_{\mathbb{K}}(V, W)$ in the finite-dimensional case. Also a very important exercise for you to make sure you do is to show that $- \otimes V$ is an exact functor. **Reminder you have a take-home quiz due via Canvas on Friday. There are two take-home quizzes next week due via Canvas, see Canvas for details.**

3. WEEK 3

3.1. Monday, September 28, 2026. We worked on Worksheets 3.1 and 3.2, which use tensor products to build algebras generated by a vector space. On Worksheet 3.1 we first looked at tensor powers $V^{\otimes d}$ and their multilinear universal property. We then constructed the tensor algebra $T(V) = \bigoplus_{d \geq 0} V^{\otimes d}$, whose multiplication is concatenation of tensors. Every linear map from V to a $\mathbb{K}$ -algebra extends uniquely to an algebra homomorphism out of $T(V)$. In the language of last week, T is left adjoint to the forgetful functor. Imposing commutativity gives the symmetric algebra $\text{Sym}(V)$, which has the same universal property for commutative algebras. Its degree d part describes symmetric multilinear maps, and choosing a basis identifies it with a polynomial ring, so $\dim \text{Sym}^d(V) = \binom{n+d-1}{d}$. That worksheet concludes with $\text{Sym}(V \oplus W) \cong \text{Sym}(V) \otimes \text{Sym}(W)$ and a comparison of symmetric powers with duality. The natural pairing between $\text{Sym}^d(V^\vee)$ and $\text{Sym}^d(V)$ is perfect when $d!$ is invertible, but it can be identically zero in characteristic p . Divided powers repair this in every characteristic.

On Worksheet 3.2 we instead quotiented $T(V)$ by the ideal generated by all $v \otimes v$. This gives the exterior algebra $\Lambda(V)$, which describes alternating multilinear maps. Increasing wedges of basis vectors form a basis, so $\dim \Lambda^d(V) = \binom{n}{d}$. A wedge $v_1 \wedge \cdots \wedge v_d$ is nonzero exactly when the v_i are linearly independent, and the exterior powers of a linear map detect its rank. Since $\Lambda^n(V)$ is a line when $\dim(V) = n$, an endomorphism ϕ acts on it by a scalar, and we defined $\det(\phi)$ to be that scalar. With this definition multiplicativity is immediate from functoriality, and expanding a wedge recovers the permutation formula. We saw that the pairing $\det(\lambda_i(v_j))$ identifies $\Lambda^d(V^\vee)$ with $\Lambda^d(V)^\vee$ in every characteristic, unlike the symmetric case. We then interpreted cofactors as matrix entries of $\Lambda^{n-1}(\phi)$. Be sure you read about adjugate matrices and the Cayley-Hamilton theorem! **Reminder you have a take-home quiz due via Canvas on Wednesday and one take-home quiz due Friday..**

3.2. Wednesday, September 30, 2026. We worked on Worksheet 3.3, on bilinear forms, quadratic forms, and trace. A bilinear form on V is the same thing as a linear map $V \rightarrow V^\vee$. It is nondegenerate when this map is an isomorphism, so a nondegenerate form identifies V with its dual in a way that depends on the form. Changing basis replaces a Gram matrix G by $P^\top G P$. This is congruence, not similarity, so rank is an invariant but the determinant only changes by a square. We sorted out symmetric, skew-symmetric, and alternating forms,

which behave differently in characteristic 2. When $\text{char}(\mathbb{K}) \neq 2$, every symmetric form has an orthogonal basis, so symmetric forms over $\mathbb{C}$ are classified by rank. Over $\mathbb{R}$, Sylvester's law of inertia shows that the numbers of positive, negative, and zero diagonal entries are invariants. Away from characteristic 2, quadratic forms are equivalent to symmetric bilinear forms via polarization, but in characteristic 2 information is lost in both directions. Nondegenerate alternating forms have a symplectic basis in every characteristic, so their rank is even and they are classified by rank alone. The worksheet concludes by defining the trace intrinsically as the composite $\text{End}_{\mathbb{K}}(V) \cong V \otimes V^\vee \xrightarrow{\text{ev}} \mathbb{K}$. In a basis it is the sum of the diagonal entries, and it satisfies cyclicity $\text{tr}_V(\psi\phi) = \text{tr}_W(\phi\psi)$. Since $\text{tr}(\text{Id}_V) = \dim(V) \cdot 1_{\mathbb{K}}$, the trace forgets part of the dimension in characteristic p . **Reminder you have a take-home quiz due via Canvas on Wednesday and one take-home quiz due Friday..**

DEPARTMENT OF MATHEMATICS, DARTMOUTH COLLEGE, HANOVER, NH 03755

Email address: juliette.bruce@dartmouth.edu

URL: <https://www.juliettebruce.xyz>